Modern Coding Agents and KG Reasoning

Block 1 Day 5

Semester 1, 2026–2027 — Teaching Week 3

Session Overview

Session objective: Evaluate coding-agent workflows, risk controls, and where structured repository knowledge improves software changes.
Timing target (min): 70
Topic anchors: coding agent; planner; tester; reviewer; impact analysis; prompt injection; secret leakage
Padlet board: https://padlet.com/qmul/b3-d5-mjxv8jfjor5277z8

Exercises

Q1 Core Concept Check

Question type Concept → **Working mode** Pair → **Expected time** 9 min

Prompt

Explain the planner-coder-tester-reviewer workflow of a coding agent in simple language. Then add one sentence on what makes this workflow different from a chatbot that only answers questions.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 9 min
- **Deliverable:** workflow summary
- **Assessment focus:** explain what makes a coding agent different from a normal chatbot
- **Expected length:** 4 role bullets + 1 sentence

How to submit

Post one pair Padlet comment listing the four roles and one short difference statement.

Discussion in class

Which role disappears first if the system becomes unsafe and overconfident?

Why this helps

Useful for short questions on coding-agent architecture and acting versus chatting.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343

- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://openai.com/codex/

Q2 Structured Problem

Question type Problem → **Working mode** Individual → **Expected time** 10 min

Prompt

A coding agent has been asked to fix a payment bug. Before it edits any code, what context should it gather? List six items drawn from the lecture, such as task goal, relevant files, tests, policies, dependencies, or likely impact area.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 10 min
- **Deliverable:** context checklist
- **Assessment focus:** identify the minimum repository context a coding agent should gather before editing
- **Expected length:** 6 items

How to submit

Post your own Padlet comment as a six-item checklist.

Discussion in class

Which missing context item is most likely to cause a wrong edit?

Why this helps

Direct practice for questions on context collection and responsible agent action.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://openai.com/codex/
- https://docs.anthropic.com/en/docs/claude-code/overview

Q3 Structured Problem

Question type Problem → **Working mode** Individual → **Expected time** 12 min

Prompt

For an enterprise payment-service repository, score these four risks from 1-5 on likelihood and impact: prompt injection from repository text, secret leakage, unsafe command execution, and passing tests with the wrong fix. Then say which one you would mitigate first.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 12 min
- **Deliverable:** risk table
- **Assessment focus:** prioritise repository risks for a coding-agent workflow
- **Expected length:** 4 rows

How to submit

Post your own Padlet comment as a four-row table plus one short priority statement.

Discussion in class

Which risk looks small at first but can become catastrophic later?

Why this helps

Direct practice for safety-risk prioritisation questions involving coding agents.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://genai.owasp.org/resource/agentic-ai-threats-and-mitigations/

Q4 Collaborative Mini-Case

Question type Case → **Working mode Group** → **Expected time 12 min**

Prompt

Design a safe pull-request workflow for a payment bug fix using planner, coder, tester, and reviewer agents. Your workflow must include one repository-memory or impact-analysis step and one human approval gate before merge.

Task details

- **Type:** case
- **Mode:** group
- **Time:** 12 min
- **Deliverable:** PR workflow design
- **Assessment focus:** design a safe pull-request workflow with impact analysis and a human gate
- **Expected length:** 6 steps

How to submit

Post one group Padlet comment with the role order, the memory or impact-analysis step, and the human gate.

Discussion in class

Where should the human reviewer intervene if only one checkpoint is allowed?

Why this helps

Supports case questions on safe orchestration of coding agents inside real software workflows.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://openai.com/codex/
- https://docs.replit.com/replitai/agent

Q5 Exam-Style Constructed Response

Question type Exam → **Working mode** Pair → **Expected time** 15 min

Prompt

Argue for a minimum safe operating standard for coding agents in production repositories. Your answer must include approval, tool limits, and audit trail.

Task details

- **Type:** exam
- **Mode:** pair
- **Time:** 15 min
- **Deliverable:** justified essay
- **Assessment focus:** argue for a minimum safe operating standard for coding agents
- **Expected length:** 180-240 words

How to submit

Post one pair Padlet comment with a clear standard, the reasons behind it, and one example of what it prevents.

Discussion in class

Which safety rule would product teams be most tempted to remove?

Why this helps

Matches long-form exam questions on governance, trust, and operational safeguards for coding agents.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://genai.owasp.org/resource/agentic-ai-threats-and-mitigations/

Q6 Stretch Extension

Question type Stretch → **Working mode** Group → **Expected time** 12 min

Prompt

Build a tiny repository graph for a payment bug fix. Include at least five nodes such as service, endpoint, database table, test suite, and policy file, then draw or describe the links and explain how this graph would help impact analysis.

Task details

- **Type:** stretch
- **Mode:** group
- **Time:** 12 min
- **Deliverable:** tiny repository graph
- **Assessment focus:** show where structured repository knowledge helps impact analysis
- **Expected length:** 5 nodes + 4 links + 1 explanation

How to submit

Post one group Padlet comment with the nodes, the links, and one short explanation of how the graph reduces risk.

Discussion in class

Which edge in your repository graph would be easiest to miss but most dangerous to ignore?

Why this helps

Good extension for synthesis questions on why structured repository knowledge helps coding agents.

References

- Russell & Norvig (2021) AIMA 4e, Ch.10, pp.314-343
- Poole & Mackworth (2023) 3e, Ch.16-17, pp.701-766
- https://qwenlm.github.io/qwen-code-docs/en/users/overview/
- https://docs.z.ai/guides/llm/glm-5.1

Submission Instructions

Use the seeded Padlet posts for Q1–Q6. Q2 and Q3 are individual. Q1 and Q5 are pair tasks. Q4 and Q6 should be one joint group comment. Start each comment with your names or group label.

Whole-Class Debrief Prompts

- What context should a coding agent never skip before editing code?
- Which risk deserves the earliest hard guardrail?
- Where does a small repository graph help more than a flat file list?

Exam Transfer Note (Day-Level)

Maps to exam questions on coding-agent workflow, context collection, risk assessment, safe PR orchestration, governance, and graph-supported impact analysis.